



UGANDA CHRISTIAN  
UNIVERSITY

A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY

Department of Computing and Technology

ADVENT 2026 SEMESTER JS ASSIGNMENT 1

---

PROGRAM: *INFORMATION TECHNOLOGY*

YEAR: 1

SEMESTER: 2

COURSE NAME: *WEB AND MOBILE APPLICATION DEVELOPMENT*

ASSIGNMENT DATE: 5<sup>th</sup> OCT 2026

DATE OF SUBMISSION: 7<sup>th</sup> Oct 2026 11:59PM

---

Develop a small console-based application of your choice using JavaScript and Node.js. The application can solve any practical problem or simulate any simple system that you find interesting.

You are not required to use a specific example. Be creative. The main objective is to demonstrate your understanding and application of functions, conditions, and loops.

Possible Ideas

- Student grading system
- Simple banking system
- Shopping or billing system

These are only suggestions. You may develop any other idea of your choice.

### Minimum Requirements

#### 1. Functions

Your program must contain and use at least 3 functions. Each function should perform a meaningful task.

Example: `function calculateTotal() { ... }`

#### 2. Conditions

Your program must use conditional statements such as if, else if, and/or else. Conditions should be used to make decisions within your application.

#### 3. Loops

Your program must use at least one loop. You may use for, while, or do...while. The loop should perform a meaningful task.

#### 4. User Interaction

The program should run in the Node.js console. Where appropriate, allow the user to provide input and interact with your application. You may use Node.js's readline module for user input.

## 5. Meaningful Output

Your application should provide clear and understandable output. The user should be able to understand what the program does, what information they need to enter, and what the program has calculated or decided.

### Code Documentation

Your code must contain comments explaining the program.

#### A. What the code does

Include comments explaining important sections of your program and the purpose of significant functions or blocks of code.

#### B. How to use the program

At the beginning of your JavaScript file, include comments explaining how the application works and how to run and use it.

```
/*  
    STUDENT GRADING SYSTEM  
  
    Description:  
    This program allows a Lecturer to enter student marks  
    and calculates the corresponding grades.  
  
    How to run:  
    1. Open a terminal.  
    2. Navigate to the folder containing this file.  
    3. Run: node student_grading.js  
  
    How to use:  
    Follow the instructions displayed in the terminal.  
*/
```

### Submission Criteria

- Submit ONE .js file containing your access number e.g. **A93044.js**.
- The program must run using Node.js without errors.
- Demonstrate at least 3 functions.
- Demonstrate conditional statements.
- Demonstrate at least one loop.
- Provide meaningful output.
- Contain comments explaining what the code does.
- Contain comments explaining how to run and use the program.

### Assessment - 30 Marks

| Criteria                               | Marks |
|----------------------------------------|-------|
| Appropriate and functional application | 5     |
| Use of functions                       | 6     |
| Use of conditions                      | 5     |
| Use of loops                           | 5     |
| User interaction and output            | 4     |
| Code comments/documentation            | 3     |
| Code organization/readability          | 2     |
| TOTAL                                  | 30    |

### Important

There is no single correct application for this assignment. The goal is not to copy a particular example, but to demonstrate that you can identify a problem, design a solution, and use functions, conditions, and loops to implement that solution in JavaScript.